SkillNavora

MODULE 12 HANDBOOK

Module 12: Automation Studio

4 sections, 28 topics explained, plus the detailed lessons, examples, exercises and official links.

Salesforce Marketing Cloud (SFMC) Academy. Verify product names, editions, licences and certifications on Salesforce's official pages.

Varun Kumar · SkillNavora · skillnavora.in
Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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Module 12: Automation Studio

Level: Intermediate. Automations: SQL activities, file imports, scheduling and monitoring.

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Detailed lessons

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12.1 Activities and scheduling

WHAT IS IT?

> An automation is a sequence of activities such as SQL Query, Data Extract, File Transfer, Import File, Filter, Script, Verification and send activities, run on a schedule or on file arrival.

WHY IS IT IMPORTANT?

> Automations prepare audiences and keep data current without manual work.

HOW DOES IT WORK AND THE STEPS

1. Create the automation, add a SQL Query Activity, then a Data Extract if needed.
2. Set a schedule (daily at 02:00 for example).
3. Run once manually before scheduling.

REAL-WORLD EXAMPLE

> Daily: import file, run audience query, filter, then send.

COMMON ERRORS

Scheduling before testing the SQL; failures then happen unattended.

INTERVIEW QUESTION

How would you schedule a daily audience build?

HANDS-ON EXERCISE

Plan an automation with four activities and an order.

OFFICIAL DOCUMENTATION AND FURTHER READING

- Salesforce Help: Marketing Cloud Engagement

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12.2 Monitoring, errors and retries

WHAT IS IT?

> Automation status shows which activity failed. Verification activities check files or row counts before the next step. Failure notifications and logs help support teams.

WHY IS IT IMPORTANT?

> Production failures need fast detection and a clear path back to working data.

HOW DOES IT WORK AND THE STEPS

1. Add a verification step after each import.
2. Set failure notifications to the owner.
3. Keep a rerun procedure.

REAL-WORLD EXAMPLE

> A verification step stops the send if the file has fewer than 1,000 rows.

COMMON ERRORS

No row-count checks, so an empty file sends an empty campaign.

INTERVIEW QUESTION

How do you troubleshoot an automation failure?

HANDS-ON EXERCISE

Write a runbook for a failed daily import.

OFFICIAL DOCUMENTATION AND FURTHER READING

- Salesforce Developer Documentation

12.1 Automation Studio Fundamentals: every topic explained

Automation Studio interface. The studio to build, schedule and monitor automations.

Automation architecture. A starting source (schedule or file drop) and steps of activities.

Automation creation. Create an automation and drag activities into steps.

Scheduling. Run hourly, daily, weekly or on file drop.

Automation monitoring. Status and run history for each automation and activity.

12.2 Automation Activities: every topic explained

SQL Query Activity. Runs SQL and writes the result to a target data extension.

Data Extract Activity. Exports data extensions or tracking data to files.

File Transfer Activity. Moves files to or from the SFTP, and decrypts or unzips them.

Import Activity. Loads a file into a data extension.

Filter Activity. Applies a saved filter to create a filtered audience.

Script Activity. Runs server-side JavaScript for advanced data work.

Verification Activity. Checks row counts and stops the automation if a rule fails.

Send activities. Send emails or SMS as a step.

Wait activities. Pause between steps.

12.3 Automation Design: every topic explained

Sequential activities. Steps run one after another; activities in the same step run together.

Scheduled automations. Automations that run on a timetable.

File-triggered workflows. Automations that start when a file arrives in a folder.

Data refresh workflows. Daily or hourly refresh of data from source files.

Audience preparation. Building send audiences with SQL and filters.

Data synchronisation. Keeping SFMC data in line with source systems.

Campaign preparation. Preparing data and sending in one flow.

12.4 Advanced Automation: every topic explained

Multiple automation dependencies. Several automations that must run in order; coordinate schedules or file drops.

Error handling. Verification steps and notifications on failure.

Retry logic. Rerun failed steps once the cause is fixed.

Automation logging. Write run results to a log data extension.

Monitoring. Watch run history and failure alerts daily.

Failure notifications. Email notifications to owners when runs fail.

Automation optimisation. Reduce steps, tune queries and avoid peak times.